

# Maximum likelihood



*10-701 Introduction to Machine Learning*  
*Geoff Gordon*

***MOO!***



- We've described several ML methods so far by giving *model, objective, optimizer*
- Often three good questions to ask about a new method — maybe not complete, but a good place to start



# ***MOO examples***



- Majority
  - ▶ model = constant functions
  - ▶ objective = minimize sum of squared errors
  - ▶ optimizer = exact (compute the mean)

# ***MOO examples***



- ID3
  - ▶ model = decision trees
  - ▶ objective = high mutual information, low depth, high number of examples per leaf node
  - ▶ optimizer = greedy top-down induction followed by reduced error pruning



# ***MOO examples***



- Gzip:
  - ▶ model: codebook tree + duplicate strings
  - ▶ objective: minimize bit-length
  - ▶ optimizer: [Deflate](#)

# ***Probabilistic machine learning***

- A recipe for making models and objectives
- Start from *generative story* of our data
  - ▶ e.g., first step  $x^{(i)} \sim N(\mu, I)$  where  $\mu \in \mathbb{R}^d$
  - ▶ ... then second step  $y^{(i)} \sim N(w^\top x^{(i)} + b, 0.1^2)$
- Each story leads to a different model
  - ▶ maybe instead  $x_j^{(i)} \sim \text{Bernoulli}(\pi_j)$  so  $x^{(i)} \in \{0,1\}^d$
  - ▶ maybe instead  $x^{(i)} \sim$  an arbitrary unknown dist'n
  - ▶ maybe instead  $y^{(i)} \sim \text{Bernoulli}(\sigma(w^\top x^{(i)} + b))$
- Each step in the story has *parameters* (e.g.,  $\mu, \pi_j, w, b$ , the arbitrary distribution of  $x^{(i)}$ )
  - ▶ our goal is to learn (some of) the parameters

# Likelihood

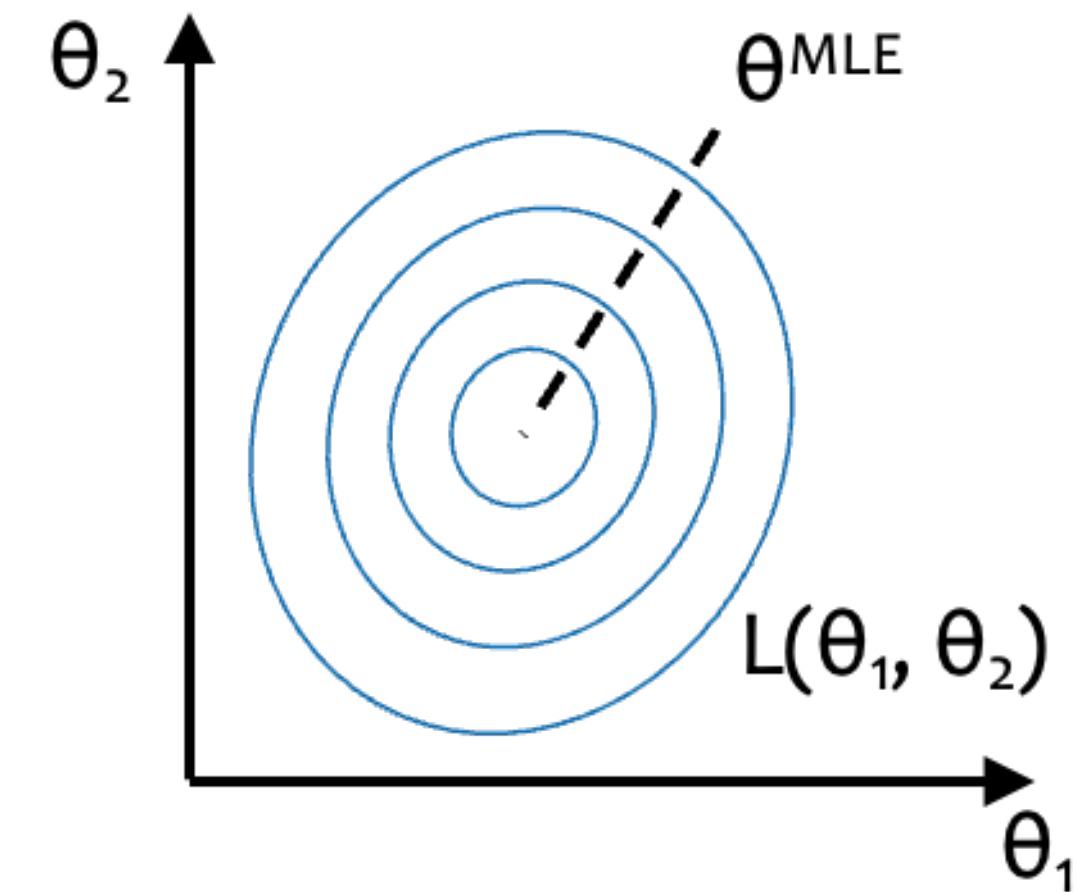
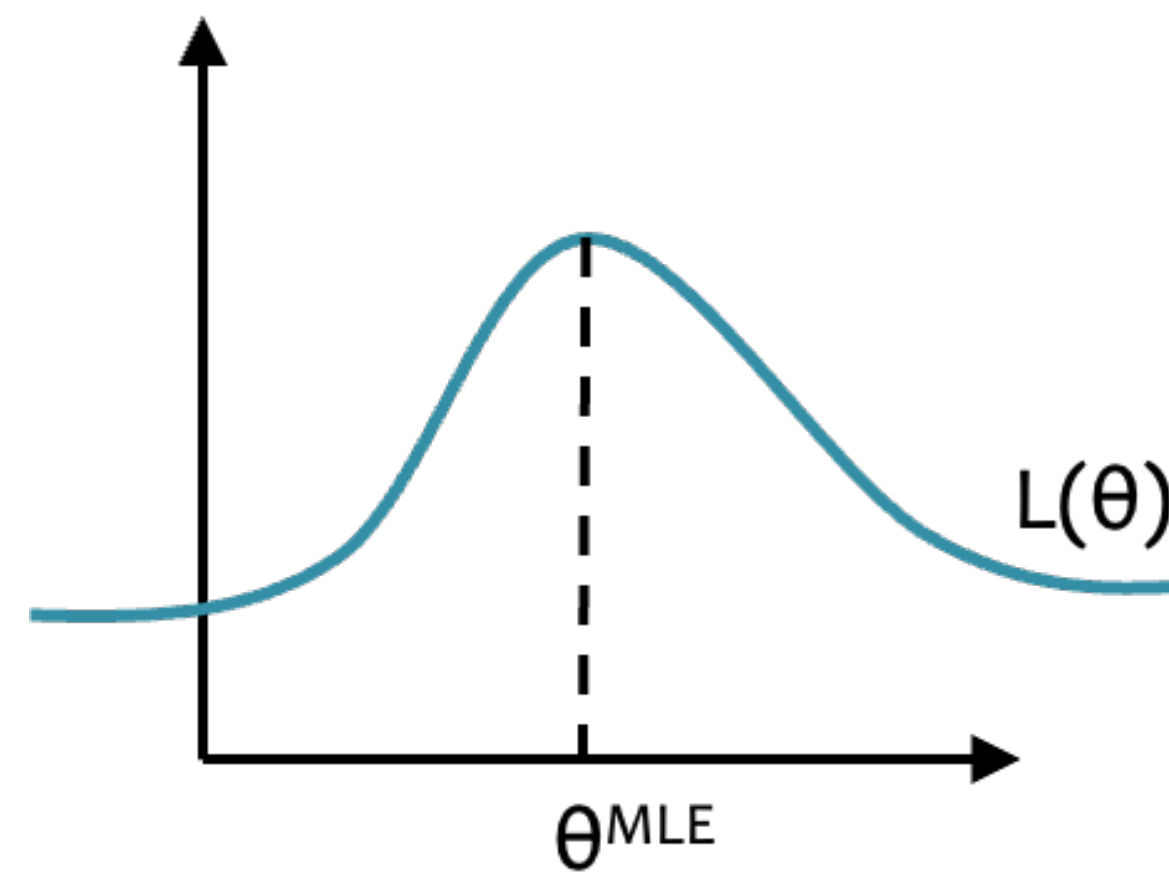
- Collect a training dataset, and use generative story to write out *likelihood*:  $P(\text{data} \mid \text{parameters})$ , considered as a function of **parameters** (hold data fixed)
  - ▶ or maybe *conditional likelihood*:  
 $P(\text{some data} \mid \text{rest of data, parameters})$
  - ▶ useful if it lets us skip learning some of the parameters
- E.g., if  $x^{(t)} \sim N(\mu, I)$  (with no  $y^{(t)}$ ),  
likelihood  $P_{\mu}(x^{(1:t)}) = \prod_{t=1}^N \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\|x^{(t)} - \mu\|^2}$
- E.g., if  $y^{(t)} \sim \text{Bernoulli}(\sigma(w^{\top}x^{(t)} + b))$ ,  
condl. lik.  $P_{w,b}(y^{(1:t)} \mid x^{(1:t)}) = \prod_{t=1}^N p_t^{y^{(t)}} (1 - p_t)^{(1-y^{(t)})}$   
where  $p_t = \sigma(w^{\top}x^{(t)} + b)$



# *Estimation*

Principle of *Maximum Likelihood Estimation*:  
Choose the parameters that maximize the  
likelihood of the data

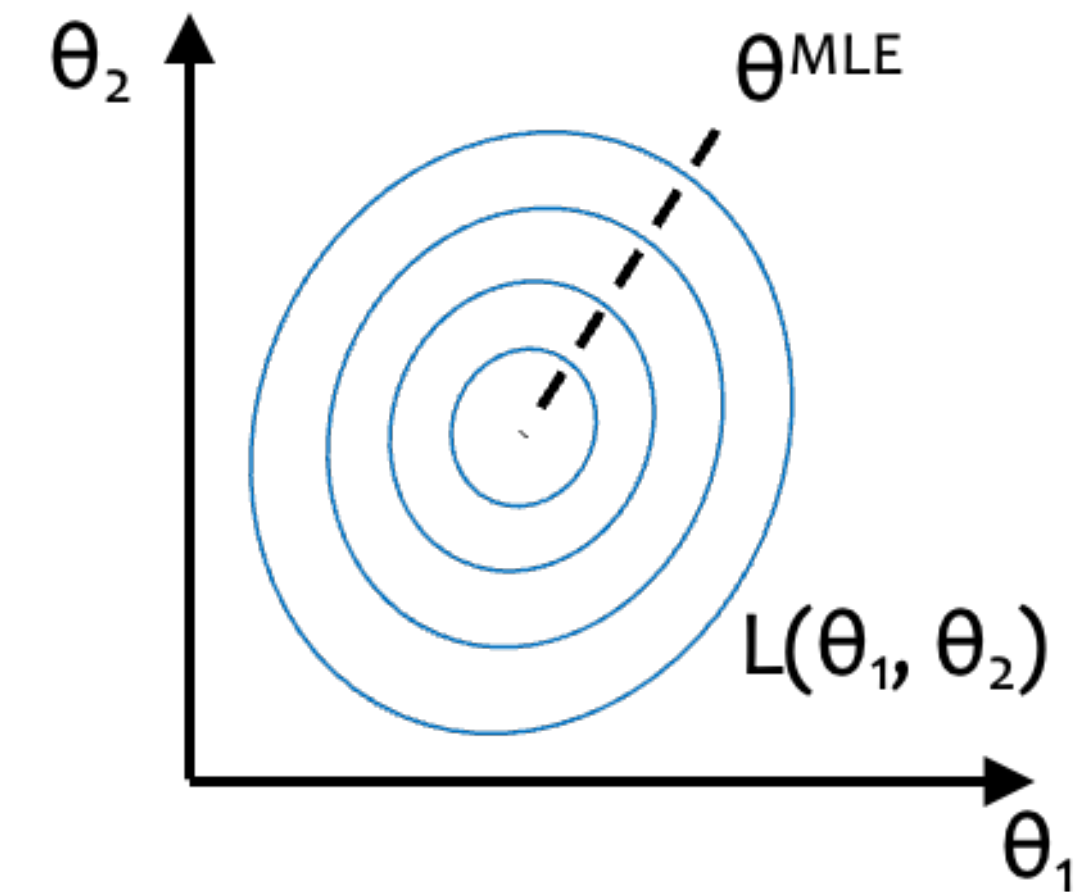
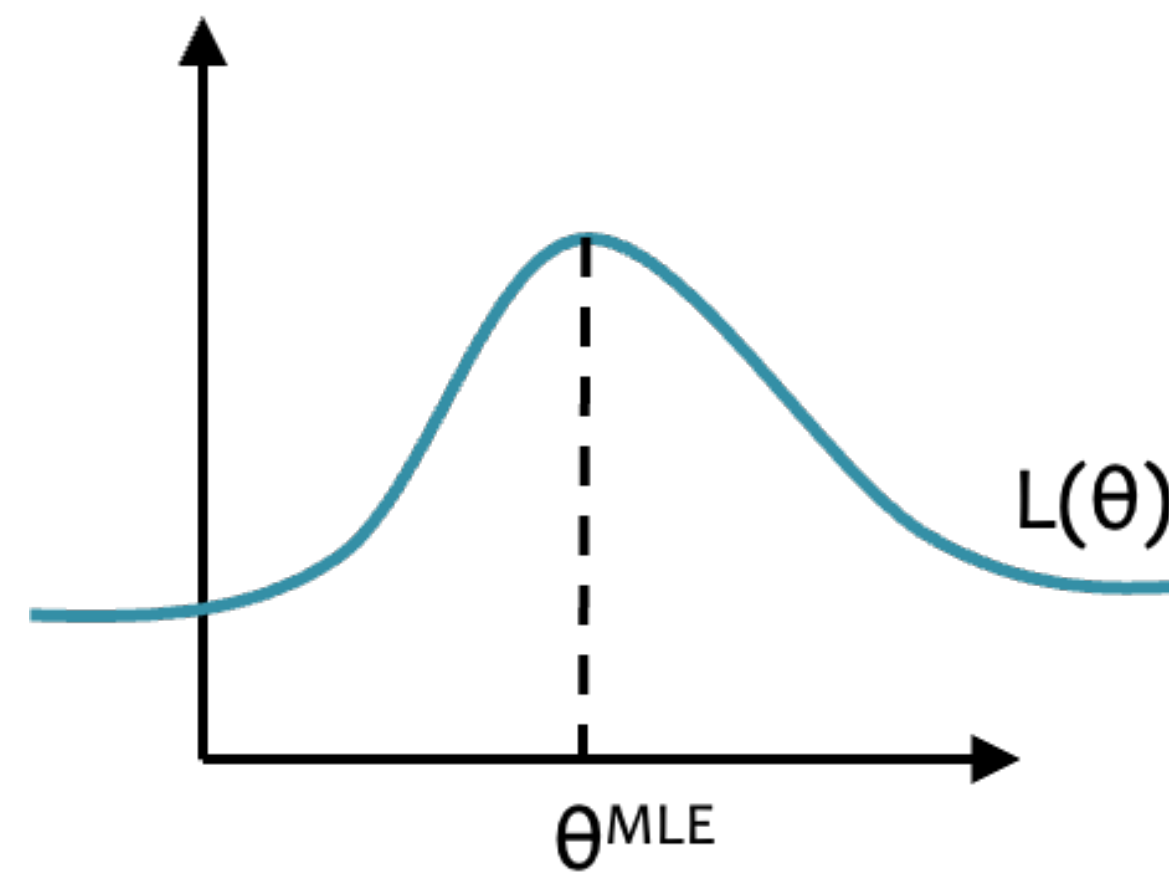
$$\theta^{\text{MLE}} = \arg \max_{\theta} \prod_{t=1}^N P(\mathbf{x}^{(t)} \mid \theta)$$



# *Estimation*

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$$\theta^{\text{MLE}} = \arg \max_{\theta} \prod_{t=1}^N P(\mathbf{x}^{(t)} \mid \mathbf{y}^{(t)}, \theta)$$

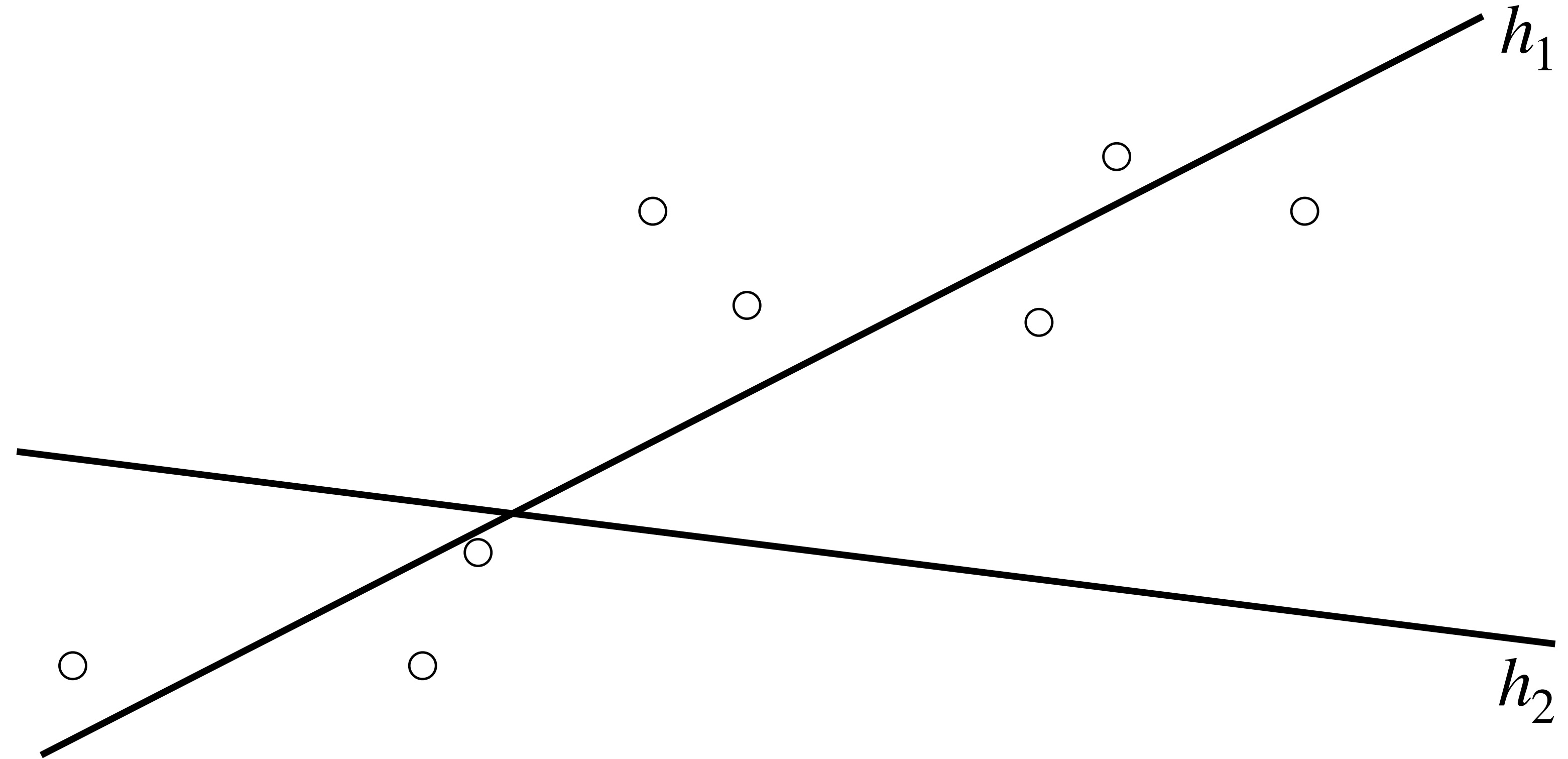


# ***MLE***

- Both are often written as ***MLE***
- Choose the model that *best explains the observed data*



***Which hypothesis is better?***



- $\exists$  only a fixed amount of total probability (sums to one)
- MLE pushes as much probability as possible to things we've observed
  - ▶ ...and steals it from things we didn't observe

***For example***

- Data: coin flips, HTT
- Model: coin lands heads w.p.  $\theta$ , iid over flips
- MLE:



# *MOO*



- Coin flips:
  - ▶ model: iid with probability  $\theta$
  - ▶ objective: log likelihood
  - ▶ optimizer: exact (differentiate and set to 0)



# *Probability reminders*



- Three common ways to describe probability:
  - ▶ probability distribution
  - ▶ probability density
  - ▶ measure



# Distribution

- *Idea: assign numbers to atomic events so that, if we assign (say) 0.23, the event happens 23% of the time*
- A probability distribution function  $p$  maps discrete atomic events to real numbers, with
  - ▶  $p(a) \geq 0$  for all atomic events  $a$
  - ▶  $\sum_{a \in U} p(a) = 1$  summed over universe  $U$  (entire table)
- E.g., uniform on  $n$  atoms:  $p(a) = \frac{1}{n}$  for each
- Notation:  $p_X, p_Y, p_{X,Z}$  — but often skip subscripts

$P(X, Y, Z)$

| $Z \rightarrow$<br>$X, Y \downarrow$ | 0   | 1   |
|--------------------------------------|-----|-----|
| 00                                   | 0.1 | 0.2 |
| 01                                   | 0.2 | 0.0 |
| 10                                   | 0.1 | 0.3 |
| 11                                   | 0.1 | 0.0 |

*also called pmf*

atomic event = smallest  
division of outcomes  
e.g.,  = 2, not  
  $\in \{\text{even \#s}\}$

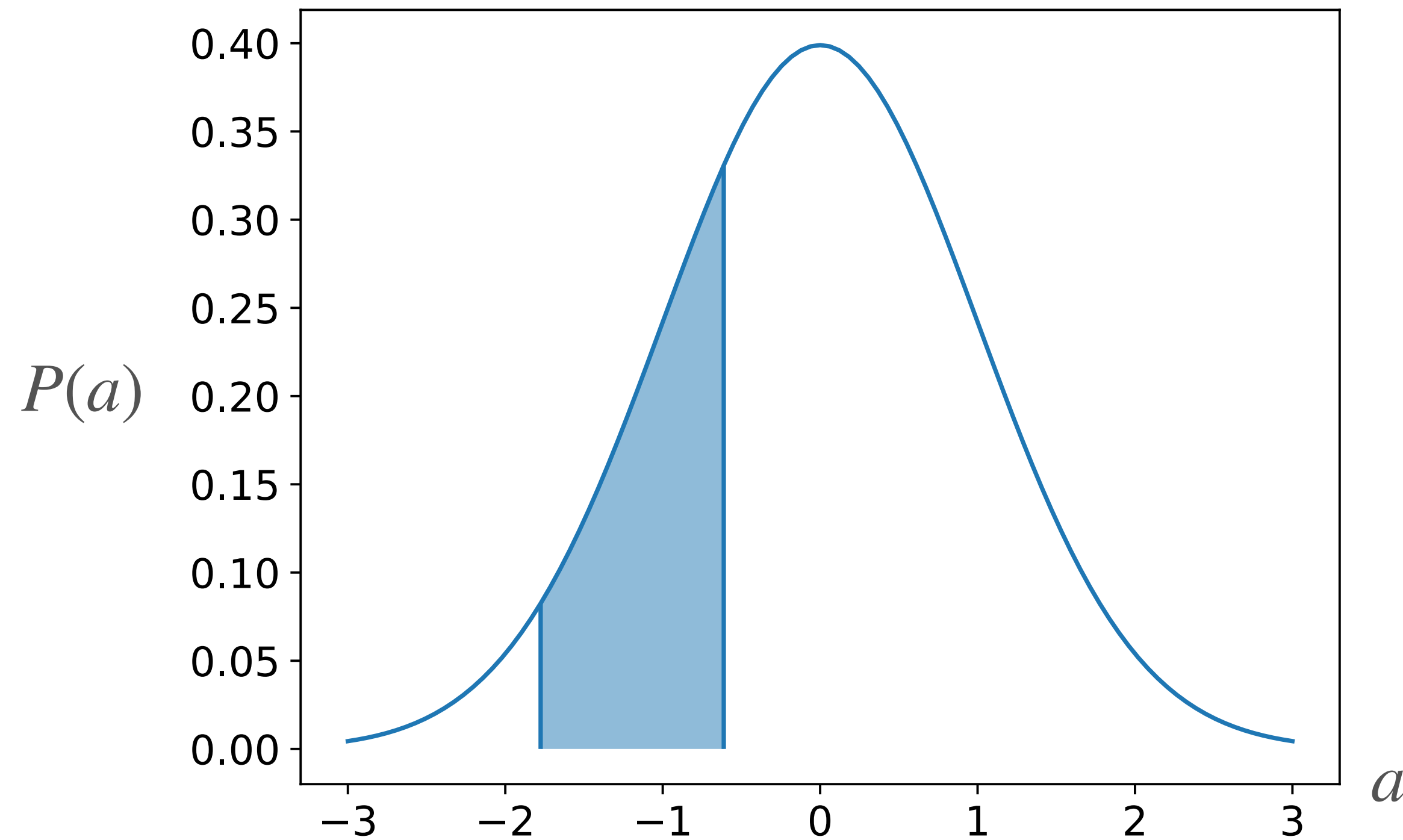
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|                   |          |          |
|-------------------|----------|----------|
| $P(X, Y, Z)$      |          |          |
| $Z \rightarrow$   | <b>0</b> | <b>1</b> |
| $X, Y \downarrow$ |          |          |
| <b>00</b>         | 0.1      | 0.2      |
| <b>01</b>         | 0.2      | 0.0      |
| <b>10</b>         | 0.1      | 0.3      |
| <b>11</b>         | 0.1      | 0.0      |

also called pmf

# Density



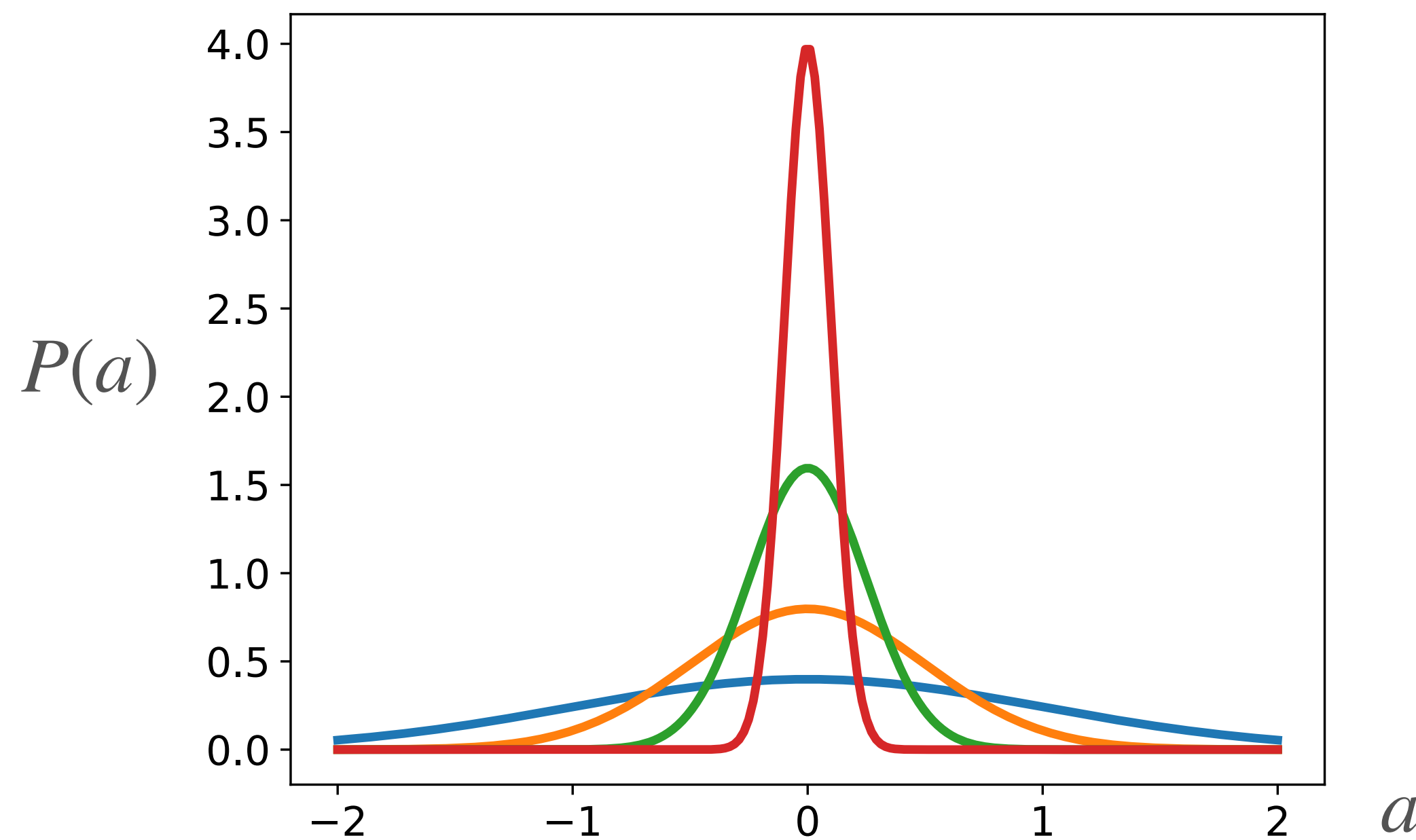
- *Idea: assign numbers to atomic events, so that integrating over atomic events yields a probability*
- A probability density function  $p$  maps continuous atomic events to real numbers, with
  - ▶  $p(a) \geq 0$  for all atomic events  $a$
  - ▶  $\int_{a \in U} p(a) da = 1$  over universe  $U$



# PDF

- Some statisticians thought they were clever making PDF be the acronym for both
  - ▶ and we use  $p(\cdot)$  or  $P(\cdot)$  for either
  - ▶ and say distribution even if we mean density (or measure)
  - ▶ worse, can see different uses in the same equation (!)
- Makes some sense since a lot of the same rules happen to apply to both representations
- But they are *not the same thing!*

***Not the  
same thing***



- E.g., for a density,  $p(0.37) \neq$  probability of seeing 0.37
  - ▶ in fact, probability of any atomic event = 0!
- E.g., for a discrete distribution,  $p(a) \leq 1$  for each outcome (atomic event)  $a$ , but for a continuous density,  $p(a)$  can be arbitrarily large
- Density acts like the *derivative* of probability

$$p(E) = \int_{a \in E} p(a) da \text{ for an event } E$$

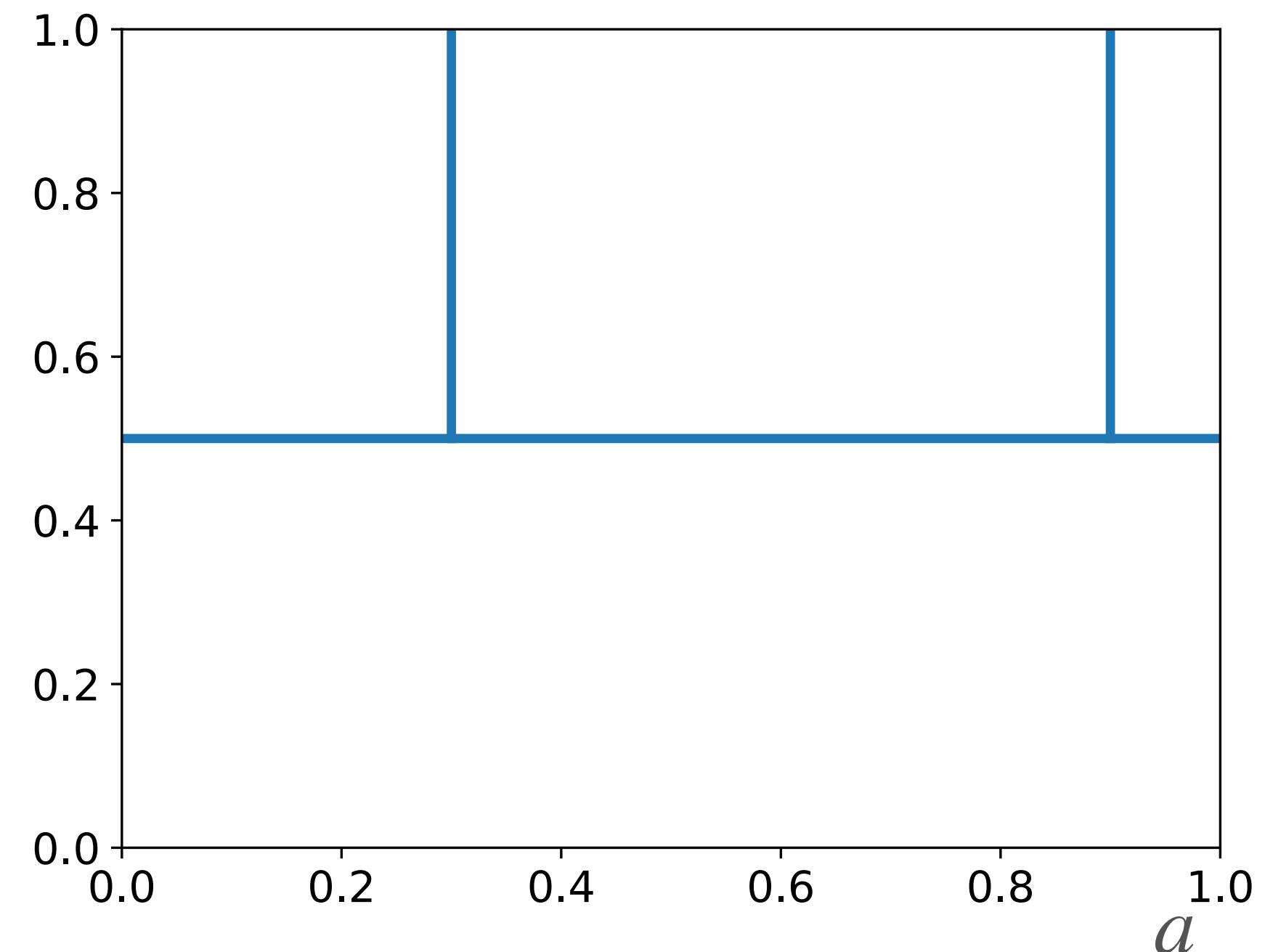




# Measure

won't typically use this  
in 10-701

$P(a)$



- *Idea: directly define the result of integration, enforcing only rules of nonnegativity, subadditivity, and  $P(U) = 1$*
- Generalizes both discrete and continuous PDFs
- Notation is like continuous:  $p(E) = \int_{a \in E} \mu(a) da$
- But OK to have atomic events with  $p(a) > 0$ 
  - ▶ imagine a Gaussian bump that's infinitely narrow centered at  $a$  (for physicists among us: a Dirac delta)
  - ▶ e.g.,  $p(x) = 0.5 + 0.25 \delta(x - 0.3) + 0.25 \delta(x - 0.9)$   
 $x \in [0, 1]$



# Conditional probability

- Describe w/ *conditional* distribution, density, or measure
  - ▶ same as above, except  $p(\cdot)$  takes conditioning event as another input,  $p(X, Y \mid Z = z)$
  - ▶ sums or integrates to 1 over  $X, Y$  for *each* atomic event  $z$  (but not other way)
- Can think of  $p(X, Y \mid Z = z)$  as a function of just  $x, y$  (with  $z$  fixed and compiled in), or as a function of both  $x, y$  and  $z$ 
  - ▶ to emphasize latter, write  $p(X, Y \mid Z)$

| $P(X, Y \mid Z)$                     |     |     |
|--------------------------------------|-----|-----|
| $Z \rightarrow$<br>$X, Y \downarrow$ | 0   | 1   |
| 00                                   | 0.1 | 0.5 |
| 01                                   | 0.2 | 0.3 |
| 10                                   | 0.6 | 0.2 |
| 11                                   | 0.1 | 0.0 |



# Factorization

| $P(X)$        | $P(Z \mid X)$   |               |                   |
|---------------|-----------------|---------------|-------------------|
| $\frac{1}{3}$ | $\frac{1}{2}$   | $\frac{1}{2}$ | $\uparrow$<br>$X$ |
| $\frac{2}{3}$ | $\frac{2}{3}$   | $\frac{1}{3}$ |                   |
|               | $Z \rightarrow$ |               |                   |

- Given joint distribution (or density or measure)  $P(X, Y, Z)$
- We can always factor it in any order:
  - ▶  $P(X \mid Y, Z) P(Y \mid Z) P(Z)$  [order  $X, Y, Z$ ]
  - ▶  $P(Y \mid X, Z) P(Z \mid X) P(X)$  [order  $Y, Z, X$ ]
  - ▶ ...
- How to interpret the above expressions?
  - ▶ discrete case: each term such as  $P(Y \mid Z)$  is a table (# outcomes for  $Y \times$  # outcomes for  $Z$ )
  - ▶ continuous case: each term such as  $P(Y \mid Z)$  is a function (arguments  $Y, Z$  — continuous analog of a table)
  - ▶ product = *broadcast* and *multiply corresponding* entries

# ***Bayes rule***

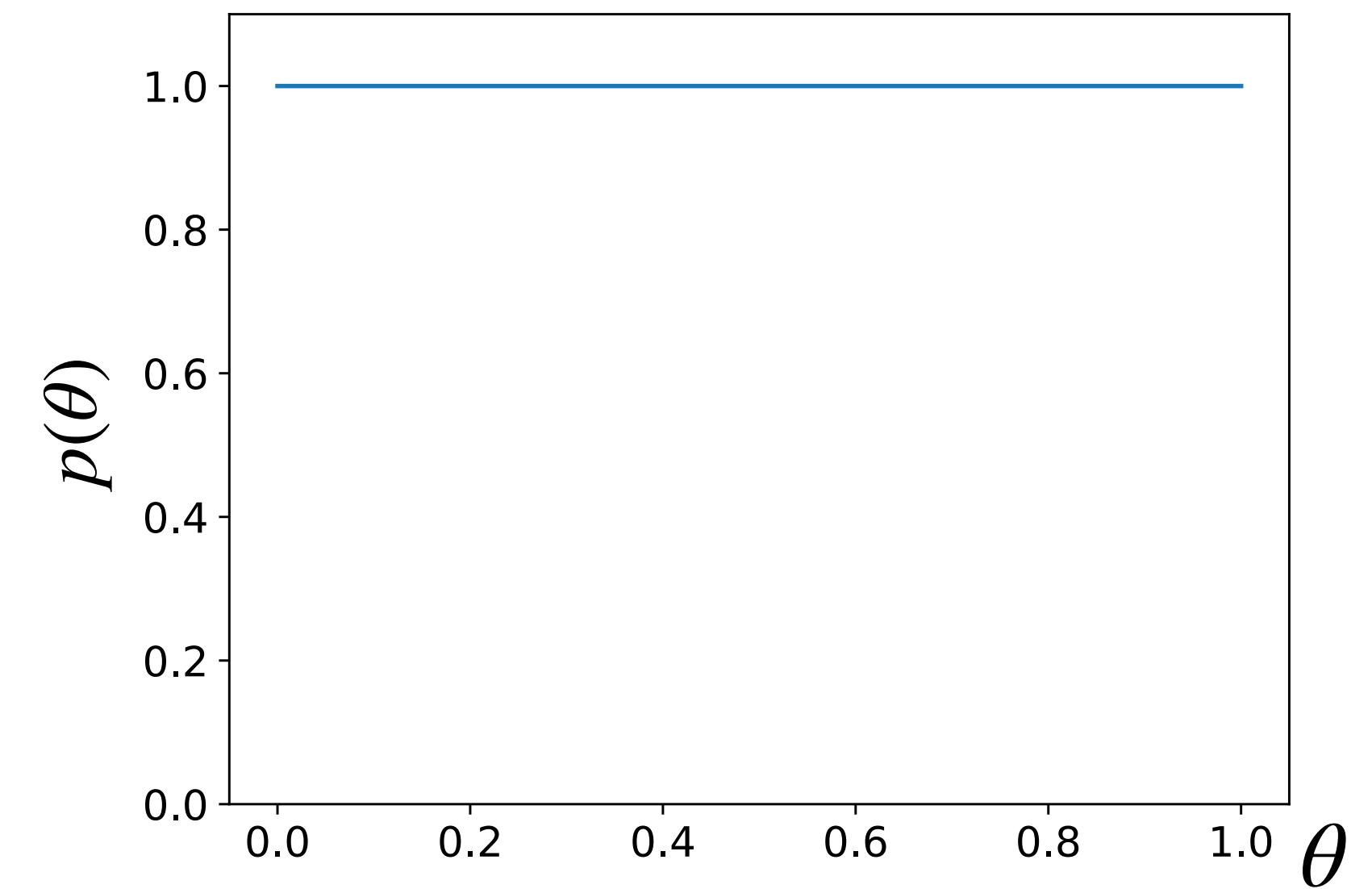
- By factorization:
  - ▶  $p(X) p(Y | X) = p(X, Y) = p(Y) p(X | Y)$
- Divide through by  $p(X)$ :
  - ▶  $p(Y | X) = p(X | Y) p(Y) / p(X) \quad \leftarrow \text{Bayes rule}$

# ***Working with probabilities***

- Should know how to work with probabilities — recitation / readings / OH / Piazza can help if you're rusty
- Discrete and continuous random variables, events; marginal, joint, and conditional probabilities; expectation and conditional expectation; mean and variance; independence and conditional independence
- Familiarity with common distributions: e.g., Gaussian, Bernoulli, multinomial
- Manipulating probability tables or continuous densities
  - ▶ including conditioning and conditional tables/densities
- Most important tools: factorization; Bayes rule; linearity of expectation
  - ▶ also broadcasting; change of variables (Jacobian rule)



# Example

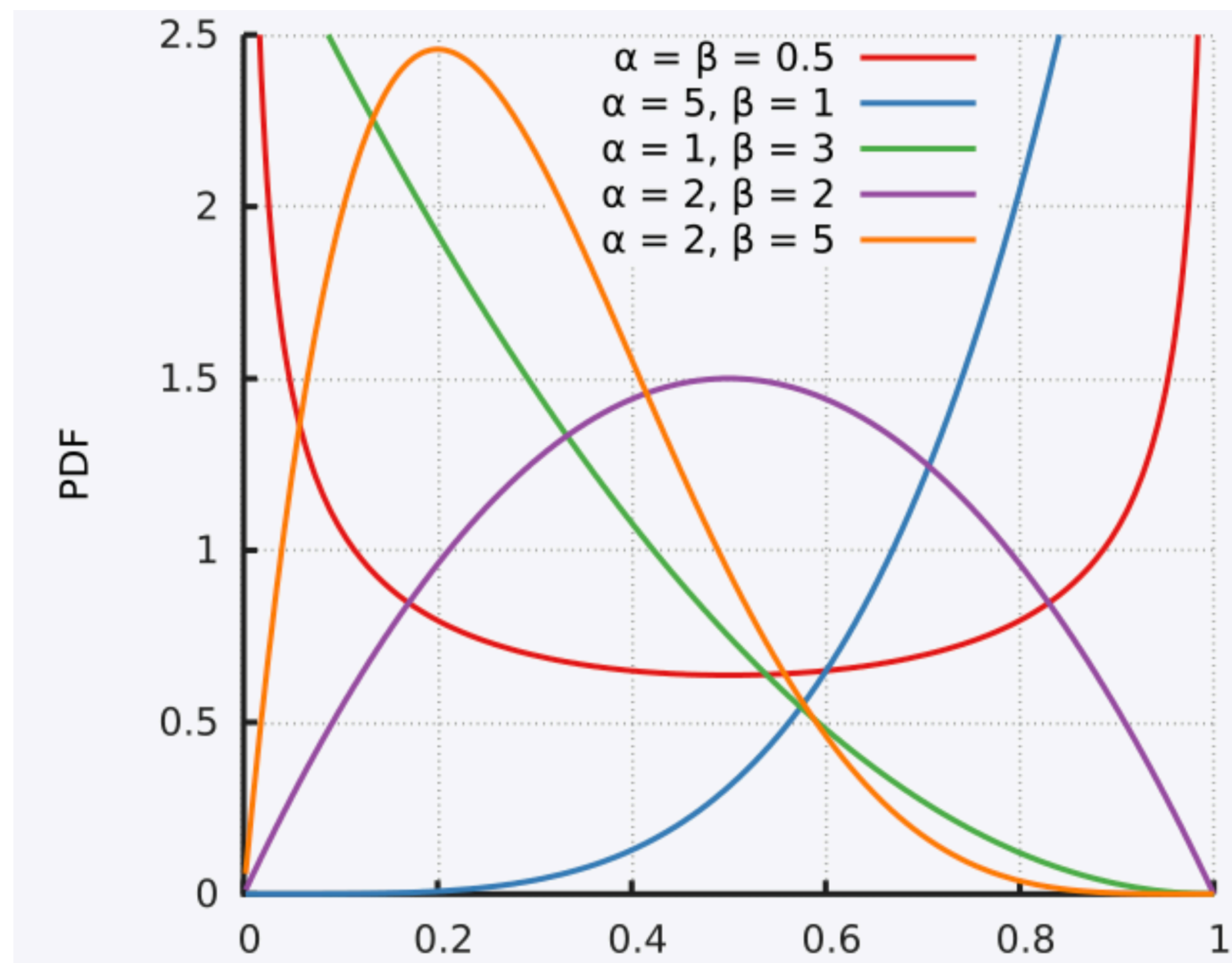


- Again, a biased coin:  
 $p(\text{heads})$  is  $\theta$
- Parameter  $\theta \in [0,1]$  selected uniformly
  - ▶ the above is what we know before seeing any data (*prior*)
- We now see 5 flips:  $X_1, X_2, \dots, X_5$  are HHTHT
- None of the flips depend on any of the others (they are conditionally independent given  $\theta$ )
- We want the distribution of  $\theta$  after seeing  $X_{1:5}$  (*posterior*)
  - ▶ terminology: prior and posterior are always relative to a given inference — often the posterior from one inference becomes the prior for the next

# ***Find posterior using Bayes rule***

- Compute prior  $\times$  likelihood, normalize
  - ▶  $P(\theta \mid X) = P(\theta) P(X \mid \theta) / P(X)$  ← *can skip denominator if we're normalizing anyway*
  - ▶ normalizer measures ***fit***: how well does our entire model class explain our observations

# Beta



plot credit: Wikipedia

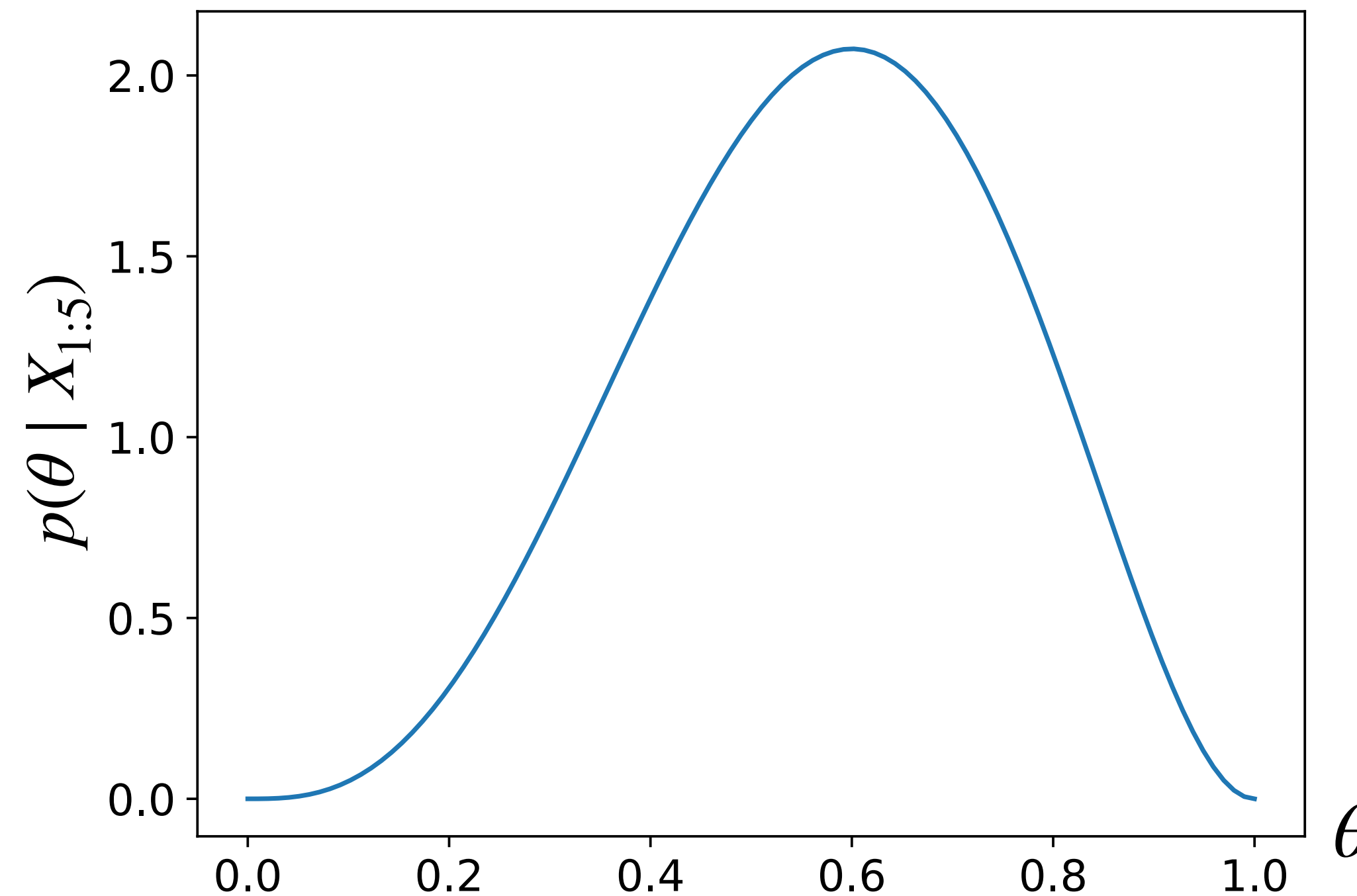
- Posterior is  $P(\theta \mid X_{1:5}) = 60 \theta^3 (1 - \theta)^2$
- This is a **Beta** distribution
  - ▶ in general,  $\propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ , parameters  $\alpha, \beta > 0$ :
    - ▶  $\alpha$  moves mass toward  $\theta = 1$ ,  $\beta$  moves it toward  $\theta = 0$
    - ▶ larger  $\alpha, \beta$  make a sharper distribution



# ***Finding the most probable parameter value***

- Posterior =  $60 \theta^3 (1 - \theta)^2$
- To find mode, differentiate and set to 0
  - ▶ becomes easier if we take log before derivative
  - ▶ log turns *product* over examples into *sum* over examples, sum rule is simpler than product rule

# MAP



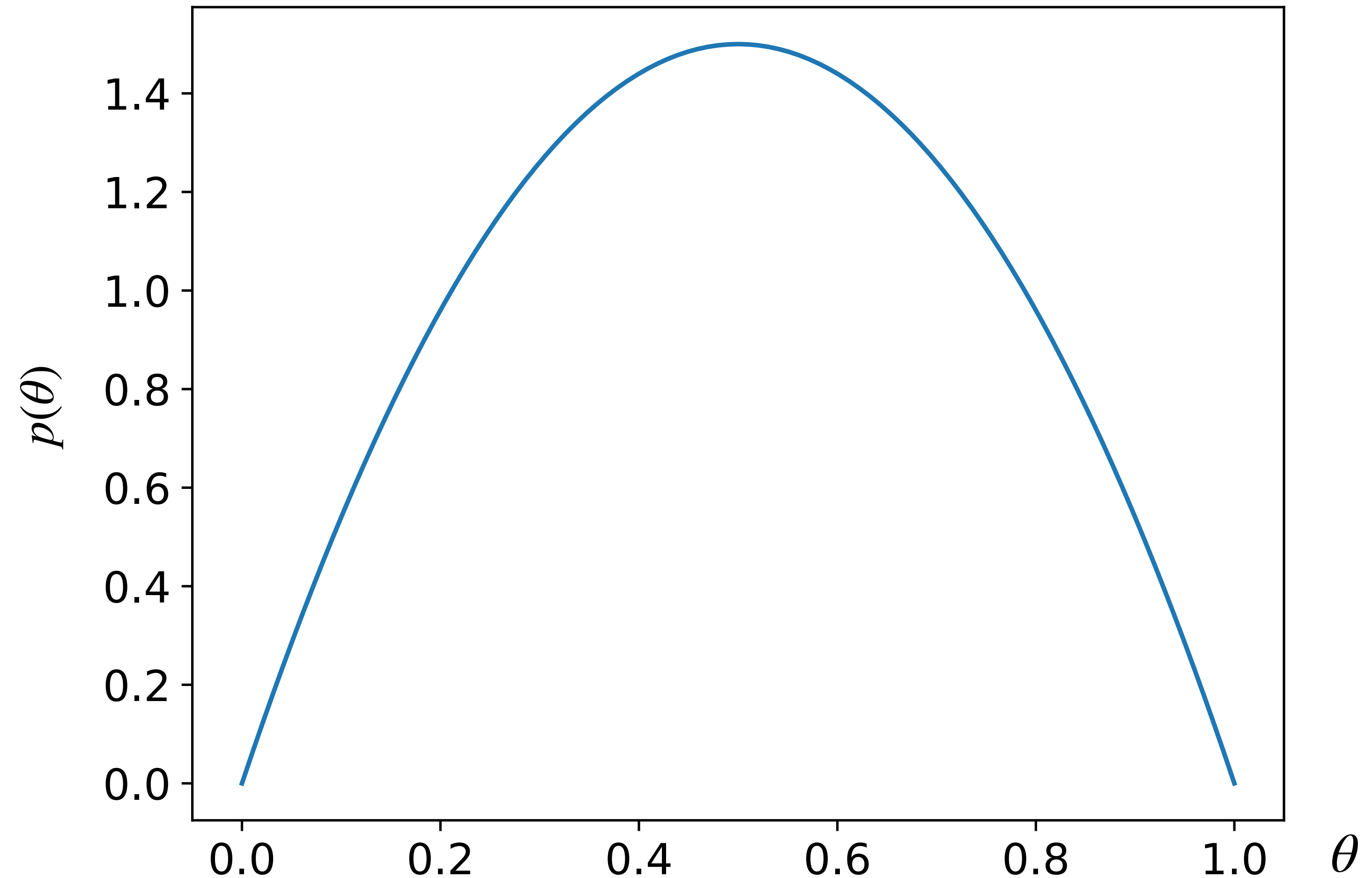
- Highest point of posterior (the *mode*) is  $\theta = \frac{3}{5}$ 
  - ▶ if we have to report a single  $\theta$ , this is a good one
  - ▶ called *maximum a posteriori* or *MAP* estimate
- Can also acknowledge uncertainty: e.g., report the whole distribution, or list of 100 samples from the distribution, or mean and standard deviation, or an interval that contains 95% of its probability

# Statistics and estimators

- Very reasonable answer:  $\frac{3}{5}$  of our data were heads, so we say  $p(H) = \frac{3}{5}$
- Observed fraction of heads is called a *statistic*: a function of our data
  - ▶ other statistics: fraction of tails;  $\ln \frac{1 + \text{\#heads}}{1 + \text{\#tails}}$
- A statistic that we use to find  $\theta$  is called an *estimator*
  - ▶ in this case,  $\mathbb{E}(\text{observed fraction}) = \text{true } \theta$
  - ▶ so our estimator is *unbiased*
- Value of estimator is *estimate* — here  $\frac{3}{5}$

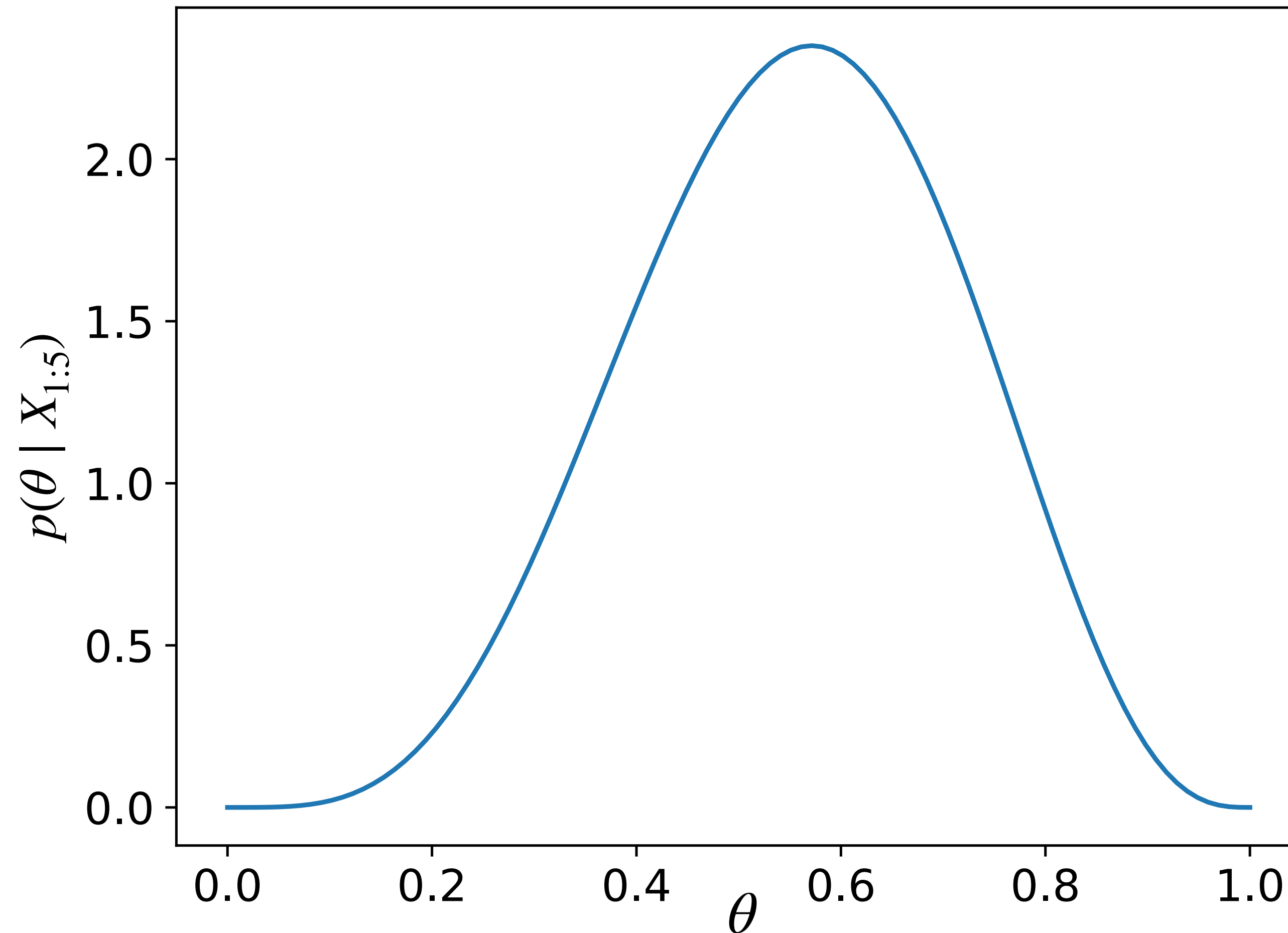


# *Changing the prior changes the estimate*



- Change prior to  $p(\theta) = 6\theta(1 - \theta)$  (a Beta distribution)
- Same data:  $X_1, X_2, \dots, X_5$  are HHTHT
- Find posterior and MAP for  $\theta$  after seeing  $X_{1:5}$

## ***New MAP***



→  $P(\theta \mid X_{1:5}) = 280 \theta^4 (1 - \theta)^3$       mode is  $\theta = \frac{4}{7}$

# ***Influence of prior***

- Same data, two priors  $\rightarrow$  two different estimates of  $\theta$ 
  - ▶ 2nd prior was narrower, pulls estimate toward center  $\frac{1}{2}$
  - ▶ uniform prior in a sense pays more attention to the data (likelihood term), doesn't favor one  $\theta$  over another



# Conjugate priors

- Something interesting happened
  - ▶ prior was uniform = Beta(1, 1), posterior was Beta(4, 3)
  - ▶ prior was Beta(2, 2), posterior was Beta(5, 4)
- Not an accident: Beta is the **conjugate prior** for Bernoulli
  - ▶ i.e., when parameter  $\theta$  has a Beta prior, and when data is Bernoulli( $\theta$ ), then posterior for  $\theta$  will also be Beta
- Happens somewhat often:
  - ▶  $x \sim N(\mu, \sigma^2)$ : conjugate prior for  $\mu$  is Normal
  - ▶  $x \sim \text{Exponential}(\lambda)$ : conjugate prior for  $\lambda$  is Gamma
- Useful since it simplifies inference:
  - ▶ e.g., observe  $h$  Heads and  $t$  Tails: add  $h$  to  $\alpha$  and  $t$  to  $\beta$



***Cf.  
maximum  
likelihood***

- Compare MAP vs. MLE:
  - ▶ in posterior  $p(\theta) p(X_{1:5} | \theta) / p(X_{1:5})$ , MAP ignored denominator (can recover by normalizing)
  - ▶ MLE also ignores prior  $p(\theta)$ , leaving only likelihood
$$\arg \max_{\theta} p(X_{1:5} | \theta)$$
$$= \arg \max_{\theta} p(X_1 | \theta) p(X_2 | \theta) \dots p(X_5 | \theta)$$
$$= \arg \max_{\theta} [\ln p(X_1 | \theta) + \dots + \ln p(X_5 | \theta)]$$
- Like setting prior to uniform, but works even when uniform prior doesn't make sense

# ***MLE vs. MAP***

- Both are *consistent*:
  - ▶ under weak assumptions, parameter  $\rightarrow$  true value as  $|\text{data}| \rightarrow \infty$
- Different advantages:
  - ▶ MLE: no need to think about prior
  - ▶ MAP: a reasonable prior can often make estimates more reliably accurate by reducing variance
    - ▶ helps w/ low data
    - ▶ which can be a problem even in huge datasets (!)

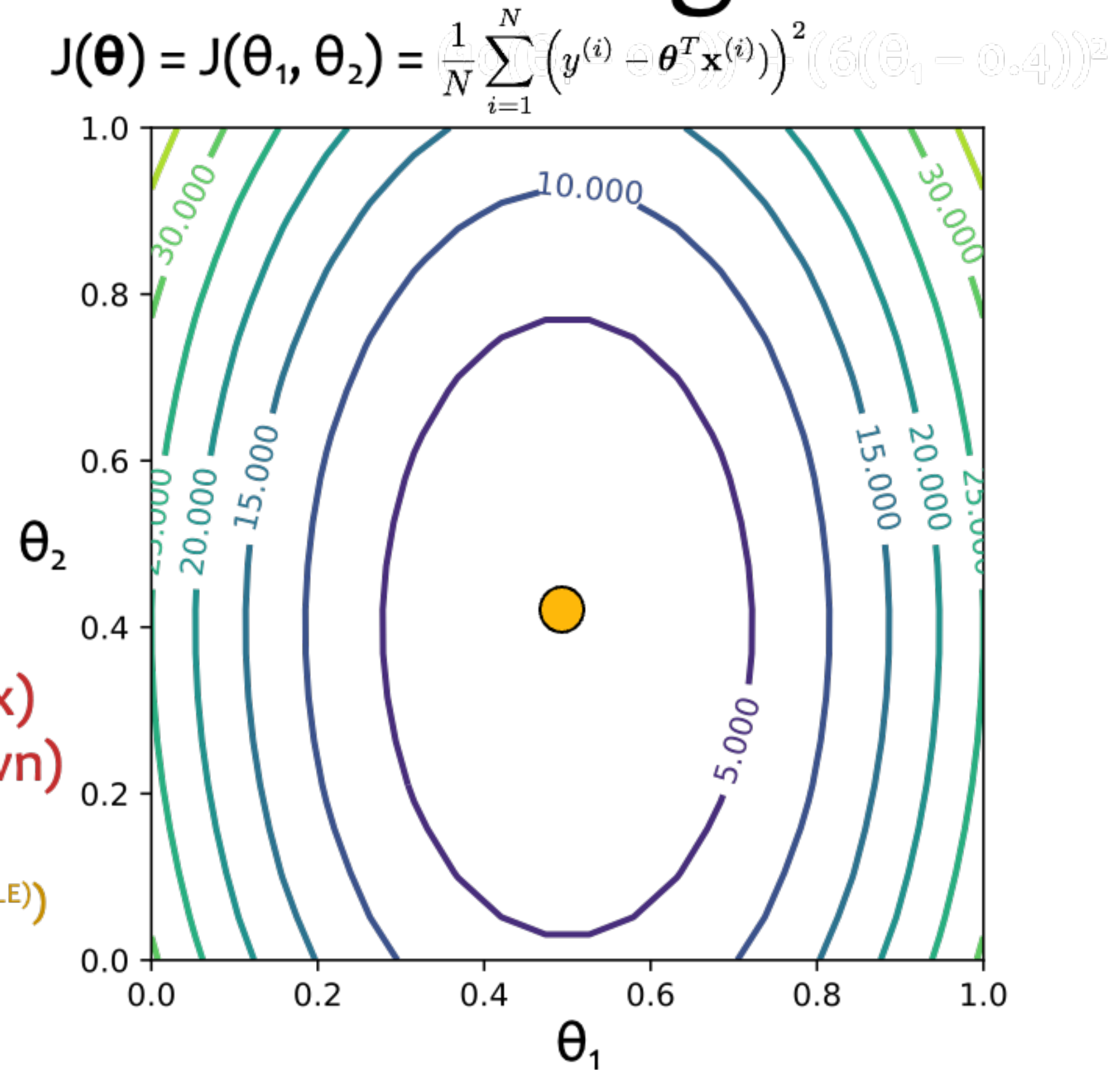
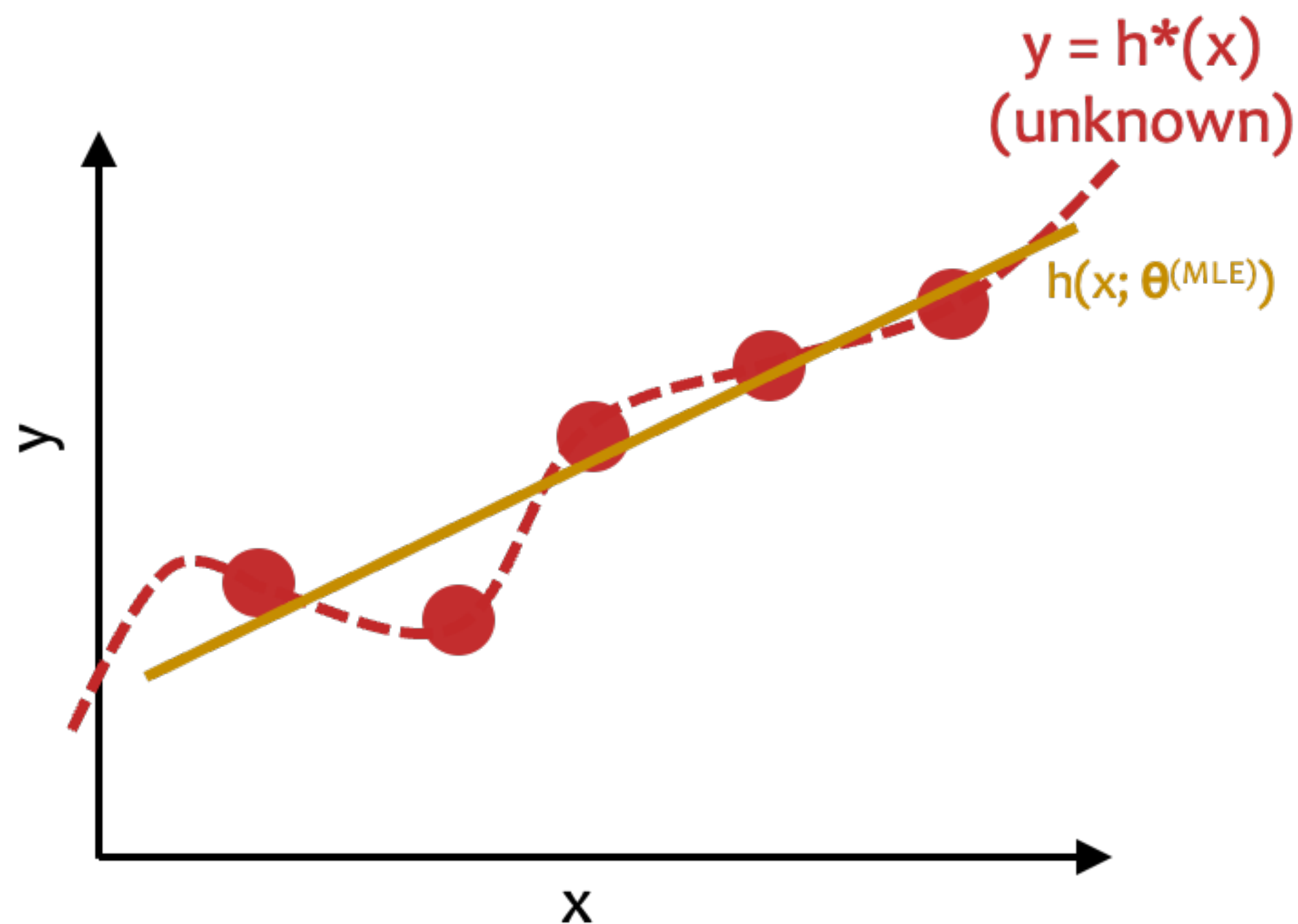
# MLE for Linear Regression

## Optimization Method #2: Closed Form

1. Evaluate

$$\theta^{\text{MLE}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

2. Return  $\theta^{\text{MLE}}$



| t   | $\theta_1$ | $\theta_2$ | $J(\theta_1, \theta_2)$ |
|-----|------------|------------|-------------------------|
| MLE | 0.59       | 0.43       | 0.2                     |

1. You'll work through the view of linear regression as a probabilistic model in the homework!
2. You'll also see how L1 and L2 regularization is equivalent to MAP estimation



# ***Philosophy of probability***

- Bayesian view (explanations so far):
  - ▶ parameters like  $\theta$  are random variables, just like data
  - ▶ prior and posterior describe our state of knowledge about  $\theta$
- Frequentist view (also very reasonable):
  - ▶ parameters like  $\theta$  are constants (properties of the world) not random variables
  - ▶ doesn't make sense to ask for a prior or posterior distribution over  $\theta$  since it's not random
- $\exists$  other philosophies of probability besides frequentist and Bayesian, but these are the two best known



# ***To MAP or not to MAP?***

- MAP doesn't make sense for frequentists! (no prior or posterior)
  - ▶ MLE does, and so does regularized MLE
  - ▶ regularized MLE is mathematically equivalent to MAP if regularizer =  $\log(\text{prior})$ , but justification is different
  - ▶ and MLE can use regularizers that aren't  $\log(\text{prior})$
- Frequentist's justifications for either:
  - ▶ consistency, convergence rates, robustness
  - ▶ e.g., find estimator that  $\rightarrow \theta$  quickly, even for worst case true distribution of data (minimax convergence rate)

# ***Difference of opinion***

- Bayesian critique of frequentist:
  - ▶ you are ignoring useful information (your incoming prior knowledge), so you are leaving performance on the table
- Frequentist critique of Bayesian:
  - ▶ you don't get the best possible minimax convergence rates, so you are leaving performance on the table
- Both are right
  - ▶ optimize different objectives → disagree on best answer
- ML'ers are typically very pragmatic: use Bayesian or frequentist reasoning depending on which leads to a practical computational solution

# ***Learning objectives— MLE, MAP***

- How can we define a probabilistic ML model using its generative story?
- What are likelihood and conditional likelihood?
- What is MLE? Why would we expect it to find a good estimate of our model's parameters?
- Find an MLE by differentiating log likelihood
- Work with discrete and continuous distributions, including differences between probability and density, as well as Bayes rule and tools on “working with probabilities” slide
- What is a MAP estimate? Find one by differentiating log prior + log likelihood, or by working with conjugate prior
- How do MLE and MAP compare?
- What are the differences between Bayesian and frequentist reasoning?